

Public Key Crypto notes - 03

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1 Public Key Encryption recap

Definition 1.1 (Public key encryption scheme).

A Public Key Encryption (PKE) scheme is a triple $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, where:

1. *Gen*: Input: security parameter 1^n (in other words, an n length string of 1s), output: (pk, sk) (public key, secret key)
2. *Enc*: Input: pk, m , output: $c \leftarrow \text{Enc}_{pk}(m)$, where $\leftarrow$ denotes that *Enc* is not necessarily deterministic
3. *Dec*: Input: sk, c , output: m or $\perp$, where $\perp$ is an error indicating an invalid ciphertext

2 Security of a Public Key Encryption scheme

- Model: The adversary A is a PPT algorithm and A knows all the public parameters (the *Gen*, *Enc*, *Dec* algorithms and the pk public key)
- The adversary A eavesdrops ciphertext c
- Goal of A is to reveal *any* information on message m
- If this goal is accomplished by A , the scheme is considered broken

Remark 2.1. This is called the CPA (Chosen-Plaintext Attack) model.

Definition 2.1 (CPA experiment).

1. Run the key generation algorithm $\text{Gen}(1^n)$, obtaining the (pk, sk) pair.
2. A is given pk , and outputs m_0, m_1 messages.
3. Choose a random bit $b \in_R \{0, 1\}$ and $c \leftarrow \text{Enc}_{pk}(m_b)$.
4. A is given c , and outputs the guess b' .
5. If the guess $b' = b$, we say that A has succeeded and $\text{PubK}_{A, \Pi}^{cpa}(n) = 1$, otherwise, $\text{PubK}_{A, \Pi}^{cpa}(n) = 0$

Definition 2.2 (CPA-secure encryption scheme). The $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ public key encryption scheme is CPA-secure if:

$$\Pr[\text{PubK}_{A, \Pi}^{cpa}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

3 ElGamal encryption scheme

An "offline version" of the Diffie-Hellman key exchange protocol.

Definition 3.1 (ElGamal encryption scheme).

- Gen:
 - Input: 1^n
 - Output: Prime $p, q \in \mathbb{Z}_p, q$ (the order of g , which is typically a prime), $x, h = g^x$.
 - In a shorter way: $p, q, g, x, h = g^x$
 - $pk = (p, q, g, h)$
 - $sk = x$
- Enc:
 - Input: $pk, m = g^i, i = 0, \dots, q - 1 = < g >$ (alternate notation)
 - Output: $c = (g^y, m \cdot h^y)$, with $y \in_R \mathbb{Z}_q$ (y is also referred to here as the *session key*)
- Dec:
 - Input: $sk, c = (c_1, c_2)$
 - Output: $\frac{c_2}{c_1^x}$

Correctness: $\frac{c_2}{c_1^x} = \frac{m \cdot h^y}{(g^y)^x} = m \cdot \frac{g^{xy}}{g^{xy}} = m$

Theorem 3.1 (CPA-security of the ElGamal public key encryption scheme). If the Decisional Diffie-Hellman (DDH) problem is hard, then the ElGamal public key encryption scheme is CPA-secure.

Recall 3.1. DDH: Given g^x, g^y, h_3 , decide whether $h_3 = g^{xy}$

Proof. Our goal is to prove that $\forall PPT$ adversaries A :

$$\Pr[PubK_{A,\Pi}(n) = 1] = \frac{1}{2} + \text{negl}(n)$$

Assume towards contradiction that we have an adversary A . We define a distinguisher D which tries to solve the DDH problem.

We define a fake encryption scheme Π' :

- Input: $m, pk = (p, q, g, h)$
- Output: $c = (g^y, m \cdot g^z)$, with $y, z \in_R \mathbb{Z}_q$
- There is no Dec algorithm defined

We define a distinguisher D :

- Input: $h_1 = g^x, h_2 = g^y m h_3$
- D tries to determine whether $g^{xy} = h_3$
- The algorithm is as follows:

1. Under pk , run A to obtain two messages: m_0, m_1
2. Choose a random bit $b \in_R \{0, 1\}$
3. Calculate $c_1 = h_2, c_2 = m_b \cdot h_3$
4. Give (c_1, c_2) to A , obtain a guess b'

- Output:

$$D = \begin{cases} 1 & \text{if } b = b', \\ 0 & \text{otherwise.} \end{cases}$$

We analyse D as follows:

- Two cases:
 1. If $h_3 = g^z, z \in_R \mathbb{Z}_q$, then $(c_1, c_2) = (g^y, m_b \cdot g^z)$, which means that this is the fake encryption scheme Π'
 - $\Pr[D(g^x, g^y, g^z) = 1] = \Pr[\text{PubK}_{A, \Pi'}^{cpa}(n) = 1] = \frac{1}{2}$ as:
 - if y, z are chosen randomly, no information can be extracted from $m \cdot g^z$, i.e. $m \cdot g^z$ is also a uniformly random element of Z_q and is independent from m .
 2. If $h_3 = g^{xy}$, then $(c_1, c_2) = (g^y, m_b \cdot g^{xy})$, which means that it is the Π ElGamal encryption scheme.
 - $\Pr[D(g^x, g^y, g^{xy}) = 1] = \Pr[\text{PubK}_{A, \Pi}^{cpa}(n) = 1]$
 - We have to prove that this probability is close to $\frac{1}{2}$

Thus:

$$\left| \frac{1}{2} - \Pr[\text{PubK}_{A, \Pi}^{cpa} = 1] \right| = \tag{1}$$

$$|\Pr[D(g^x, g^y, g^z) = 1] - \Pr[D(g^x, g^y, g^{xy}) = 1]| \leq \text{negl}(n) \tag{2}$$

by the hardness assumption of DDH.

□

4 RSA

Rivest-Shamir-Adleman

Definition 4.1 (Textbook RSA).

- Gen:
 - Input: 1^n
 - Output: (N, e, d) , where e is the public key, and d is the secret key
 - Output is generated as follows:
 1. Choose two primes p, q of bitsize n
 2. $N = p \cdot q, \varphi(n)$ (Euler's totient function) $= (p-1)(q-1)$
 3. Choose e such that $\gcd(e, \varphi(n)) = 1$

4. Compute $d \equiv e^{-1} \pmod{\varphi(N)}$, i.e. d is the solution of $e \cdot x \equiv 1 \pmod{\varphi(N)}$
5. $pk = (N, e), sk = d$

- **Enc:**

- Input: $pk = (N, e), m \in \mathbb{Z}_N^*$ (where $\mathbb{Z}_N^* = \{1 \leq k \leq N \cdot \gcd(k, N) = 1\}$)
- Output: $c = m^e \pmod{N}$

- **Dec:**

- Input: $sk = d, c$
- Output: $c^d \pmod{N}$

Correctness:

- $c^d \equiv (m^e)^d \equiv m^{ed} \pmod{N} = m^{1+k\varphi(N)} \pmod{N}$ (by the choice of d) $= m \cdot (m^{\varphi(N)})^k \equiv m \pmod{N}$
- Where $m^{\varphi(N)} = 1$ by the Euler-Fermat theorem.

Recall 4.1 (Euler-Fermat theorem). For any N and $\gcd(a, N) = 1$:

$$a^{\varphi(N)} \equiv 1 \pmod{N}$$

Remark 4.1.

- We need $\gcd(e, \varphi(N)) = 1$ to have a solution for d .
- We required that $\gcd(m, N) = 1$, but usually, we relax this requirement, since:

$$\Pr[\gcd(m, N) \neq 1] (\text{either } p|m \text{ or } q|m) = \frac{q+p}{N} \approx \frac{1}{p} \approx 2^{-n} = \text{negl}(n)$$